Akoizo Original Practice - Form 4

Division B | Scrambler | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Wheels are 6.0 cm in diameter. How far does one axle turn move the vehicle?

A. 12.0 cm

B. 37.7 cm

C. About 18.8 cm

D. 6.0 cm

Answer: _____

2. (2 points) If vehicle speed doubles, how does its kinetic energy change?

Unit: Energy from a falling mass.
Worked-case data: A 1.50 kg mass falls 0.60 m. If 40 percent reaches a 0.50 kg vehicle as kinetic energy, what is its top speed?
Separate investigation observations:
2.0 kg mass: Top speed 4.2 m/s; egg cracked on stop in 3 of 5 runs.
1.2 kg mass: Top speed 2.6 m/s; egg safe 5 of 5; time +1.8 s.
Target: 8.00 m in both cases.

A. It halves

B. It stays the same

C. It becomes four times larger

D. It doubles

Answer: _____

3. (2 points) What must be detached when impounding?

Unit: Calibration and competition day.
Worked-case data: Table: 30 turns -> 7.20 m, 34 turns -> 8.15 m, 38 turns -> 9.10 m. Set for a State target of 8.60 m.
Separate investigation observations:
Run 1: Straight run: 4.2 cm from target, 6.1 s, no cans.
Run 1 score: 100 + 8.4 + 6.1 = 114.5.
Option for Run 2: 60 cm can gap; bonus runs average 9 cm error, 6.5 s.

A. The falling mass

B. The wheels

C. The stopping mechanism

D. The egg holder

Answer: _____

4. (2 points) What happens if the egg breaks on the first run?

Unit: Protecting the egg.
Worked-case data: A vehicle arrives at 1.5 m/s and stops in 6 cm. Find the average deceleration in g.
Separate investigation observations:
Design A: Arrives 3.0 m/s, stops in 2 cm; 2 of 5 eggs cracked.
Design B: Arrives 1.5 m/s, stops in 5 cm; 0 of 5 cracked.
Question: Compare decelerations.

A. A violation, and no second run

B. Only a time penalty

C. Nothing

D. A 300-point penalty

Answer: _____

5. (4 points) Why does a bigger distance ratio reduce acceleration?

Unit: Drivetrains: turning a drop into distance.
Worked-case data: A vehicle has 0.50 cm axle radius and 4.0 cm wheel radius. The mass pulls 0.75 m of string. About how far can the vehicle be driven?
Separate investigation observations:
Run 1: String wound in one neat layer: 8.42 m.
Run 2: String piled in three layers: 7.61 m.
Same: Mass and drop height.

Answer: _____
_____
_____

6. (4 points) What violations come from the egg and barrier?

Unit: Rules and the low-score system.
Worked-case data: A run stops 12.4 cm from the Target Point in 5.31 s with an Inside Can Distance of 60.0 cm and no violations. Find the Run Score.
Separate investigation observations:
Vehicle A: Stops 20 cm away in 3.0 s.
Vehicle B: Stops 6 cm away in 7.5 s.
Both: No cans, no violations.

Answer: _____
_____
_____

7. (4 points) Calculate the arc length for a chord of 8.00 m and a radius of 11.0 m.

Answer: _____
_____
_____

8. (4 points) How should a team find the largest error source?

Unit: Automatic stopping.
Worked-case data: Target 9.25 m. Wheels are 7.50 cm in diameter. The vehicle skids 3 cm after braking, and the egg tip sits 10 cm ahead of the point used as the start reference. How many turns?
Separate investigation observations:
Setting 30.0 turns: Stops at 7.21, 7.24, 7.19 m.
Setting 35.0 turns: Stops at 8.40, 8.43, 8.38 m.
Target: 8.00 m.

Answer: _____
_____
_____

9. (4 points) How much kinetic energy does a 0.80 kg vehicle have at 2.5 m/s?

Answer: _____
_____
_____

10. (4 points) Why test on a floor similar to the tournament's?

Unit: Calibration and competition day.
Worked-case data: Table: 30 turns -> 7.20 m, 34 turns -> 8.15 m, 38 turns -> 9.10 m. Set for a State target of 8.60 m.
Separate investigation observations:
Run 1: Straight run: 4.2 cm from target, 6.1 s, no cans.
Run 1 score: 100 + 8.4 + 6.1 = 114.5.
Option for Run 2: 60 cm can gap; bonus runs average 9 cm error, 6.5 s.

Answer: _____
_____
_____

11. (6 points) Solve this case and explain your reasoning: A vehicle arrives at 1.5 m/s and stops in 6 cm. Find the average deceleration in g.

Answer: _____
_____
_____

12. (2 points) Why did Run 2 travel less?

Supplied synthetic observations:
Run 1: String wound in one neat layer: 8.42 m.
Run 2: String piled in three layers: 7.61 m.
Same: Mass and drop height.

A. The mass was lighter

B. Pile-up increased the effective axle radius

C. The wheels shrank

Answer: _____

13. (2 points) Define “Target Point” in the context of Scrambler.

Answer: _____
_____
_____

14. (2 points) Define “Bonus Line” in the context of Scrambler.

Answer: _____
_____
_____

15. (2 points) Define “Threaded rod” in the context of Scrambler.

Answer: _____
_____
_____

16. (2 points) Define “Launcher” in the context of Scrambler.

Answer: _____
_____
_____

17. (2 points) Define “Impound” in the context of Scrambler.

Answer: _____
_____
_____

18. (2 points) Define “g-force” in the context of Scrambler.

Answer: _____
_____
_____